

PROTOTYPE BOM

Source Data From: **Softcore_SA_HW.PrjPcb**
 Project: **Softcore_SA_HW.PrjPcb**
 Variant: **None**

SoftCore SA Adapter

Company Part Number: **IMR-006-SCH**
 PCB Revision: **2**
 Engineer: **Hab S Collector**



Report Date: 3/22/2026
 Print Date: 22-Mar-26

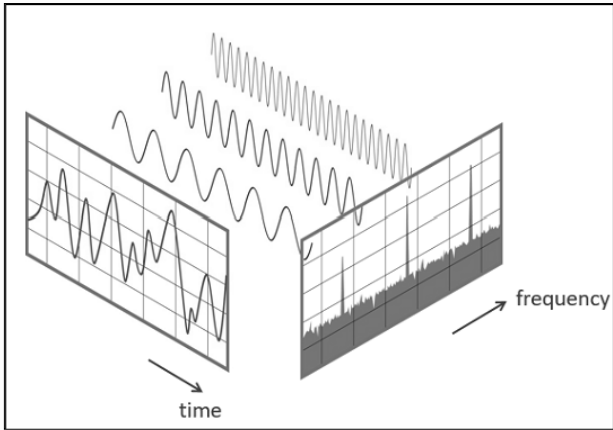
6:05:43 PM
 6:06:11 PM

#	Quantity	LibRef	Description	Designator	MFG	MFG PN	Supplier 1	Supplier Part Number 1	Unit Price	TOTAL
1	5	CAP_SMD_47UF_16V_1210	CAPCER 47UF 50V XSR 1210	C1, C4, C15, C43, C46	Murata Electronics	GRM32ER60J476ME20L	DigiKey	490-1887-1-ND	28	\$1.40
2	18	CAP_SMD_0.1UF_50V_0805	CAPCER 0.1UF 50V X7R 0805	C2, C3, C5, C6, C10, C11, C12, C13, C16, C17, C20, C22, C23, C26, C27, C32, C33, C35	KYOCERA AVX	KGM21NR71H104KT	DigiKey	478-KGM21NR71H104KTCT-ND	1	\$1.80
3	1	CAP_SMD_470PF_50V_0805	CAPCER 470PF 50V COGNP0 0805	C7	Yageo	CC0805JRNPO9BN471	DigiKey	311-1119-1-ND	11	\$0.11
4	5	CAP_SMD_10UF_50V_0805	CAPCER 10UF 50V X5R 0805	C8, C30, C40, C41, C49	Murata Electronics	GRM21BR61H100KE43L	DigiKey	490-18663-1-ND	28	\$1.40
5	2	CAP_SMD_220PF_50V_0805	CAPCER 220PF 50V COGNP0 0805	C9, C14	Yageo	CC0805JRNPO9BN221	DigiKey	311-1115-1-ND	11	\$0.22
6	2	CAP_SMD_3.3NF_50V_0805	CAPCER 3.3NF 50V COGNP0 0805	C18, C19	Murata Electronics	GRM2165C1H832JA01D	DigiKey	490-1632-1-ND	13	\$0.26
7	2	CAP_SMD_1UF_50V_0805	CAPCER 1UF 50V X7R 0805	C21, C31	KYOCERA AVX	08055C105K4T2A	DigiKey	478-KAM21KR71H105KUJCT-ND	27	\$0.54
8	2	CAP_SMD_4.7NF_50V_0805	CAPCER 4.7NF 50V X7R 0805	C28, C38	KEMET	C0805C472K5RACAUTO	DigiKey	399-6969-1-ND	12	\$0.24
9	3	CAP_SMD_100UF_16V_SMD	CAPCER 100UF 16V X7S SMD	C29, C36, C37	Taiyo Yuden	BMK325ABJ107MM-P	DigiKey	587-5426-1-ND	1.92	\$5.76
10	3	CAP_SMD_4.7UF_50V_0805	CAPCER 4.7UF 50V X5R 0805	C34, C45, C47	Murata Electronics	GRM21BR61H475KE51L	DigiKey	490-10751-1-ND	19	\$0.57
11	1	CAP_SMD_22PF_50V_0805	CAPCER 22PF 50V X7R 0805	C39	KEMET	C0805C220K5RACAUTO	DigiKey	399-17627-1-ND	24	\$0.24
12	1	CAP_SMD_470NF_50V_0805	CAPCER 0.47UF 6.3V X5R 0402	C42	Murata Electronics	GRM155R60J474KE190	DigiKey	490-3266-1-ND	10	\$0.10
13	1	CAP_SMD_27PF_50V_0805	CAPCER 27PF 50V COGNP0 0805	C44	KEMET	C0805C270J5GACTU	DigiKey	399-C0805C270J5GACTU-ND	11	\$0.11
14	1	CAP_SMD_2.2UF_50V_0805	CAPCER 2.2UF 50V X5R 0805	C48	Murata Electronics	GRM219R61E225KA12J	DigiKey	490-GRM219R61E225KA12JCT-ND	38	\$0.38
15	6	DIODE_1N4148W-7-F	RELAY REED SPDT 250MA 5V	D1, D2, D3, D4, D5, D6, D7, D8	Diodes Inc	1N4148W-7-F	DigiKey	1N4148W-FDCT-ND	16	\$1.28
16	4	CONN_HEADER_12P_100ML_RA	CONN HEADER R/A 12POS 2.54MM	J1, J3, J4, J6	Samtec Inc.	TSW-106-08-T-D-RA	Digi-Key	SAM1049-06-ND	93	\$3.72
17	1	CONN_MICRO_SD_DM8BT-DSF-PEJS	CONN MICRO SD CARD R/A SMD	J2	Hirose Electric Co Ltd	DM8BT-DSF-PEJS	DigiKey	HRL942CT-ND	2.30	\$2.30
18	1	CONN_HEADER_3P_100ML_VERT	CONN HEADER VERT 3POS 2.54MM	J5	Sullins Connector Solutions	PREC003SFAN-RC	Digi-Key	S1212EC-03-ND	07	\$0.07
19	1	CONN_MOLEX_0731713150	CONN BNC RCTP R/A 50 OHM PCB	J7	Molex	0731713150	DigiKey	WM20430-ND	1.40	\$1.40
20	4	CONN_HEADER_2P_100ML_VERT, CONN_JUMPER_2P_100ML_VERT, CONN_JUMPER_2P_100ML_VERT, CONN_JUMPER_2P_100ML_VERT	CONN HEADER VERT 2POS 2.54MM	J8, J9, J10, J12	Sullins Connector Solutions	PREC002SFAN-RC	Digi-Key	S1212EC-02-ND	07	\$0.28
21	1	CONN_BATT_18650	CONN HEADER VERT 2POS 2.54MM	J11	Keystone Electronics	54	Digi-Key	36-54-ND	1.10	\$1.10
22	1	INDUCTOR_74439324022	POWER INDUCTOR 4030; 2.2UH 7.80	L1	Würth Elektronik	74439324022	DigiKey	732-74439324022CT-ND	1.80	\$1.80
23	1	INDUCTOR_VLS3012ET-150M	FXED IND 15UH 550MA 516MOHM SMD	L2	TDK Corporation	VLS3012ET-150M	DigiKey	445-6660-1-ND	26	\$0.26
24	5	LED_GREEN_LTST-CL70GKT	LED GREEN CLEAR SMD	LED1, LED2, LED3, LED7, LED8	Lite-On	LTST-CL70GKT	DigiKey	160-1179-1-ND	10	\$0.50
25	2	LED_YELLOW_LTST-CL70YKT	LED YELLOW CLEAR SMD	LED4, LED5	Lite-On	LTST-CL70YKT	DigiKey	160-1175-1-ND	15	\$0.30
26	1	LED_RED_LTST-CL70EKT	LED RED CLEAR SMD	LED6	Lite-On	LTST-CL70EKT	DigiKey	160-1178-1-ND	18	\$0.18
27	4	MECHANICAL_SCREW_M3X6_PAN_HEAD	SCREW M3X6 PAN HEAD SS	MECH1, MECH2, MECH3, MECH4	AFM Hexseal	RM3X6MM 2701	DigiKey	335-1156-ND	58	\$2.32
28	4	MECHANICAL_STANDOFF_M3X20	HEX STANDOFF M3 BRASS 20MM	MECH5, MECH6, MECH7, MECH8	Keystone Electronics	24407	DigiKey	36-24407-ND	1.36	\$5.44
29	1	MECHANICAL_SPEAKER_MOUNTED	SPEAKER PANEL MOUNT	MECH9	ARDUINO	312815-3W	ARDUINO	312815-3W	2.50	\$2.50
30	1	MODULE_ARTY_A7	DEVELOPMENT BOARD FPGA XILINX A7	MECH10	Digilent, Inc.	410-319-1	DigiKey	1286-1206-ND	299.00	\$299.00
31	2	MOSFET_NCH_2N7002-7-F	MOSFET N-CH 60V 115MA SOT23-3	Q1, Q3	Diodes Inc	2N7002-7-F	DigiKey	2N7002-FDICT-ND	20	\$0.40
32	2	MOSFET_PCH_RLML5103TRPBF	MOSFET P-CH 30V 760MA SOT23	Q2, Q4	Infineon Technologies	RLML5103TRPBF	DigiKey	RLML5103PBFCT-ND	45	\$0.90
33	8	RES_SMD_240_0805_1/8W_1%	RES SMD 240 OHM 1% 1/8W 0805	R1, R2, R3, R4, R5, R6, R7, R8	Yageo	RC0805FR-13240RL	DigiKey	13-RC0805FR-13240RLCT-ND	1	\$0.80
34	26	RES_SMD_10K_0805_1/4W_1%	RES SMD 10K OHM 1% 1/4W 0805	R9, R10, R11, R12, R13, R14, R15, R16, R17, R18, R19, R20, R22, R28, R29, R30, R31, R32, R33, R35, R36, R38, R39, R40, R44, R49	Yageo	RC0805FR-7W10KL	DigiKey	13-RC0805FR-7W10KLCT-ND	10	\$2.60
35	1	RES_SMD_200_0805_1/8W_1%	RES SMD 200 OHM 1% 1/8W 0805	R21	Yageo	RC0805FR-07200RL	DigiKey	311-200CRCT-ND	1	\$0.10
36	1	RES_SMD_3.0K_0805_1/8W_1%	RES SMD 3.0K OHM 1% 1/8W 0805	R23	Yageo	RC0805FR-133KL	DigiKey	13-RC0805FR-133KLCT-ND	1	\$0.10
37	4	RES_SMD_1.0K_0805_1/8W_1%	RES SMD 1.0K OHM 1% 1/8W 0805	R24, R27, R41, R45	Yageo	RC0805FR-071KL	DigiKey	311-1.0KCRCT-ND	1	\$0.40
38	2	RES_SMD_1.5K_0805_1/8W_0.1%	RES SMD 1.5K OHM 0.1% 1/8W 0805	R25, R26	Yageo	RT0805BRD071K5L	DigiKey	YAG1814CT-ND	16	\$0.32
39	2	RES_SMD_22K_0805_1/8W_1%	RES SMD 22K OHM 1% 1/8W 0805	R34, R37	Yageo	RC0805FR-0722KL	DigiKey	311-22.0KCRCT-ND	10	\$0.20
40	1	RES_POT_10K_2XGANG_LINEAR	POT 10K OHM 120W LINEAR - 2X GANG	R42	Bourns Inc.	PRS12R-4115K-103B1	DigiKey	118-PRS12R-4115K-103B1-ND	2.56	\$2.56
41	1	RES_SMD_7.5K_0805_1/8W_1%	RES SMD 7.5K OHM 1% 1/8W 0805	R43	Yageo	RC0805FR-077K5L	DigiKey	311-7.50KCRCT-ND	10	\$0.10
42	2	RES_SMD_2.49K_0805_1/8W_1%	RES SMD 2.49K OHM 1% 1/8W 0805	R46, R64	Yageo	RC0805FR-132K49L	DigiKey	13-RC0805FR-132K49LCT-ND	1	\$0.20
43	6	RES_SMD_0_0805_1/8W_1%	RES SMD 0 OHM 1% 1/8W 0805	R47, R48, R54, R61, R62, R72	Yageo	RC0805JR-070RL	DigiKey	311-0.0ARCT-ND	1	\$0.60
44	1	RES_POT_10K_1XGANG_LINEAR	POT 10K OHM 120W LINEAR - 1X GANG	R50	Bourns Inc.	3386P-1-103TLF	DigiKey	3386P-103TLF-ND	2.20	\$2.20
45	1	RES_SMD_1.69K_0805_1/8W_1%	RES SMD 1.69K OHM 1% 1/8W 0805	R51	Yageo	RC0805FR-071K69L	DigiKey	311-1.69KCRCT-ND	1	\$0.10
46	1	RES_SMD_178K_0805_1/8W_1%	RES SMD 178K OHM 1% 1/8W 0805	R52	Yageo	RC0805FR-07178KL	DigiKey	311-178KCRCT-ND	10	\$0.10
47	2	RES_SMD_102K_0805_1/8W_1%	RES SMD 102K OHM 1% 1/8W 0805	R53, R56	Yageo	RC0805FR-07102KL	DigiKey	311-102KCRCT-ND	10	\$0.20
48	7	RES_SMD_100K_0805_1/4W_1%	RES SMD 100K OHM 1% 1/4W 0805	R55, R57, R63, R66, R67, R68, R70	Yageo	RC0805FR-07100KL	DigiKey	311-100KCRCT-ND	13	\$0.91
49	1	RES_SMD_48.7K_0805_1/8W_1%	RES SMD 48.7K OHM 1% 1/8W 0805	R58	Yageo	RC0805FR-0748K7L	DigiKey	311-48.7KCRCT-ND	1	\$0.10
50	1	RES_SMD_17.4K_0805_1/8W_1%	RES SMD 17.4K OHM 1% 1/8W 0805	R59	Yageo	RC0805FR-0717K4L	DigiKey	311-17.4KCRCT-ND	10	\$0.10
51	1	RES_SMD_17.8K_0805_1/8W_1%	RES SMD 17.8K OHM 1% 1/8W 0805	R60	Yageo	RC0805FR-0717K8L	DigiKey	311-17.8KCRCT-ND	10	\$0.10
52	1	RES_SMD_6.81K_0805_1/8W_1%	RES SMD 6.81K OHM 1% 1/8W 0805	R65	Yageo	RC0805FR-076K81L	DigiKey	311-6.81KCRCT-ND	10	\$0.10
53	1	RES_SMD_1M_0805_1/8W_1%	RES SMD 1M OHM 1% 1/8W 0805	R69	Yageo	RC0805FR-071ML	DigiKey	311-1.00MCRCT-ND	12	\$0.12
54	1	RES_SMD_255K_0805_1/8W_1%	RES SMD 255K OHM 1% 1/8W 0805	R71	Yageo	RC0805FR-07255KL	DigiKey	311-255KCRCT-ND	10	\$0.10
55	1	RELAY_HE712C0500	RELAY REED SPDT 250MA 5V	REL1	Littlefuse	HE712C0500	DigiKey	HE112-ND	3.77	\$3.77
56	1	CONN_SPKR_HEADER_2P_100ML	CONN SPEAKER HEADER GENERIC VERT 2POS 2.54MM	SPK1	GENERIC	TBD	Digi-Key	TBD	1.00	\$1.00
57	5	SWITCH_EVQ-Q2U02W	SWITCH TACTILE SPST-NO 0.02A 15V	SW1, SW2, SW3, SW4, SW5	American Zettler	EVQ-Q2U02W	DigiKey	P12954SCT-ND	25	\$1.25
58	1	SWITCH_SLW-12689-4A-N-D	SWITCH SLIDE SPDT 300MA 50V	SW6	Same Sky	SLW-12689-4A-N-D	DigiKey	2223-SLW-12689-4A-N-D-ND	48	\$0.48
59	8	CONN_TEST_POINT_S1751-46R	PCB TEST POINT CONDUCT	TP5, TP8, TP10, TP12, TP14, TP15, TP16, TP17	Harvi in Inc	S1751-46R	Digi-Key	952-1478-1-ND	28	\$2.24
60	2	IC_MCP23S08T-E/SO	IC IO XPRDR 10MHZ SPI 18SOIC	U1, U2	Microchip	MCP23S08T-E/SO	DigiKey	MCP23S08T-E/SOCT-ND	1.54	\$3.08
61	1	MODULE_DISPLAY_128X64_2.42	DISPLAY 128X64 SSD1309 2.42" OLED WHITE	U3	UNKNOWN	SSD1309_128X64_2.42"	Amazon	SSD1309 2.42"	16.99	\$16.99
62	2	IC_AD746ARIZ-REEL7	IC ADC 12BIT SAR SOT23-6	U4, U6	Analog Devices	AD746ARIZ-REEL7	DigiKey	AD746ARIZ-REEL7CT-ND	12.02	\$24.04
63	3	IC_OPAAMP_AD8606ARZ-REEL7	IC OPAAMP GP2 CIRCUIT 8SOIC	U5, U7, U8	Analog Devices	AD8606ARZ-REEL7	DigiKey	AD8606ARZ-REEL7CT-ND	4.94	\$14.82
64	1	IC_LM4990MMXNQB	IC AMP CLASS AB MONO 2W 8VSSOP	U9	Texas Instruments	LM4990MMXNQB	DigiKey	296-37579-1-ND	45	\$0.45
65	1	IC_OPAAMP_TLV9302DR	IC OPAAMP GP2 CIRCUIT 8SOIC	U10	Texas Instruments	TLV9302DR	DigiKey	296-53513-1-ND	66	\$0.66
66	1	IC_SMPS_TPS552892RYQR	IC REG BUCK ADJ 1A 16HTSSOP	U11	Texas Instruments	TPS552892RYQR	DigiKey	296-TPS552892RYQRCT-ND	6.42	\$6.42
67	1	IC_REG_AZ1117CH-ADJTRG1	IC REG BUCK ADJ 2A SOT23	U12	Diodes Incorporated	AZ1117CH-ADJTRG1	DigiKey	AZ1117CH-ADJTRG1DICT-ND	2	\$0.20
68	1	IC_SMPS_LM43601PWPR	IC REG BUCK ADJ 1A 16HTSSOP	U13	Texas Instruments	LM43601PWPR	DigiKey	296-40221-1-ND	4.41	\$4.41

Approved

Notes

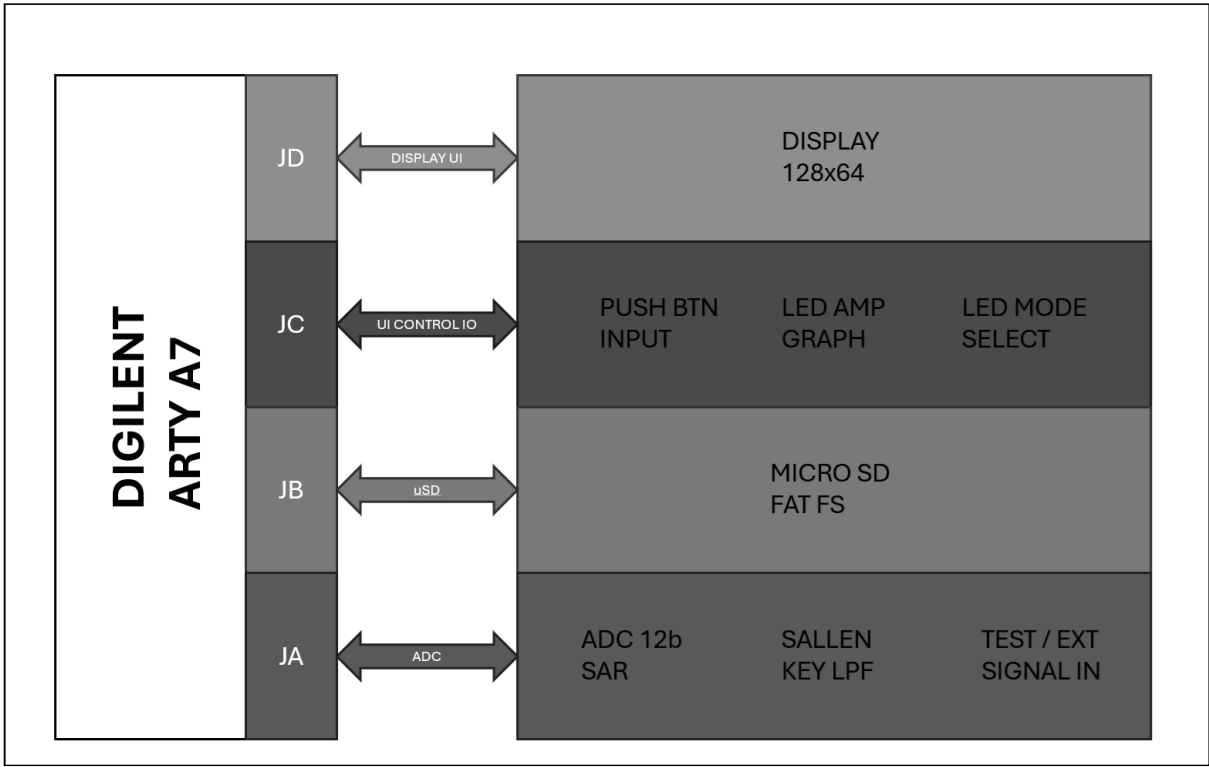
Bill of Cost (Unit Buy): \$428.80



APPLICATION

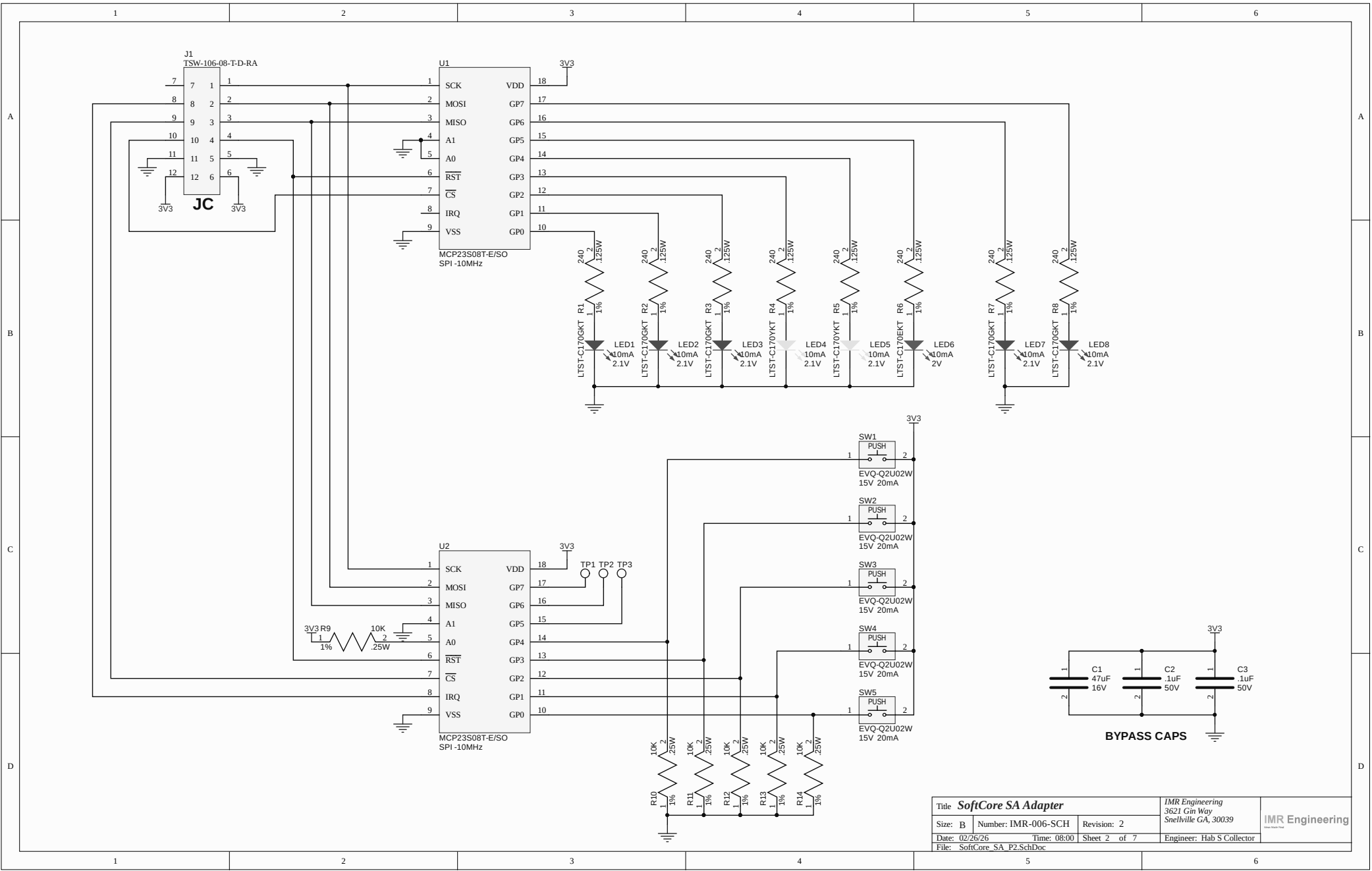
REVISION HISTORY:

REV	ECO	CHANGE DESCRIPTION
1	NA	Initial design
2	NA	Changes to REV 1schematic to make work - PCB will not be updated



HARDWARE

Title SoftCore SA Adapter			IMR Engineering 3621 Gin Way Snelville GA, 30039	
Size: B	Number: IMR-006-SCH	Revision: 2	IMR Engineering	
Date: 02/26/26	Time: 08:00	Sheet 1 of 7	Engineer: Hab S Collector	
File: SoftCore_SA_P1.SchDoc				



Title SoftCore SA Adapter			IMR Engineering 3621 Gin Way Snelville GA, 30039	
Size: B	Number: IMR-006-SCH	Revision: 2	IMR Engineering	
Date: 02/26/26	Time: 08:00	Sheet 2 of 7	Engineer: Hab S Collector	
File: SoftCore_SA_P2.SchDoc				

REV 1 TO REV 2 CORRECTIONS:
REPLACE C8 WITH SHORT CIRCUIT

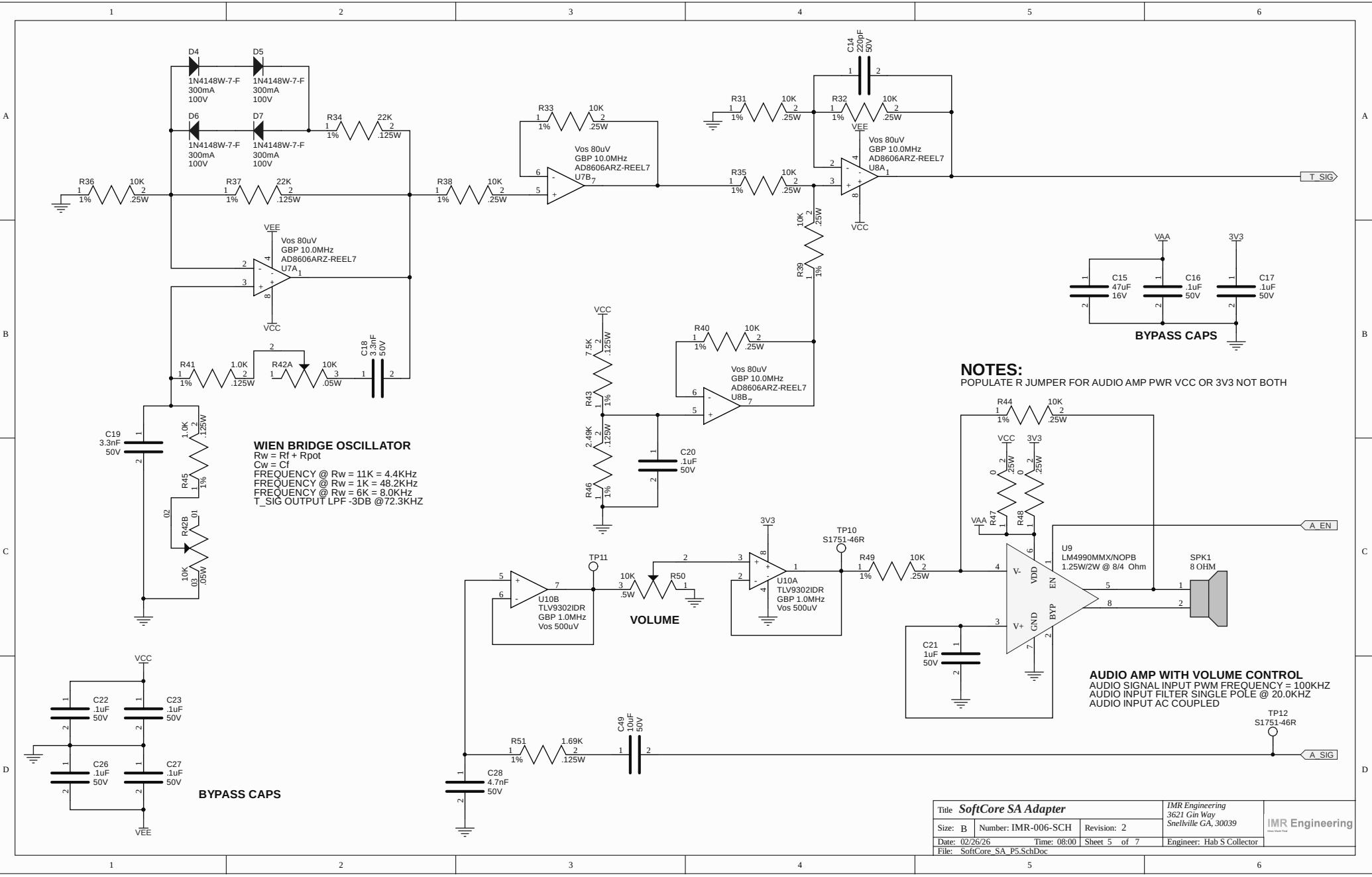
ANTI-ALIASING Sallen Key LPF
FREQUENCY CUTOFF = 329.9KHZ
ADC SAMPLE RATE TO BE 100KHZ
RX TO BALANCE +/- INPUT - NOT INCLUDED IN FILTER CALCULATION

AC COUPLING AND PROTECTION CIRCUIT

BYPASS CAPS

BNC

Title SoftCore SA Adapter			IMR Engineering 3621 Gin Way Snelville GA, 30039	
Size: B	Number: IMR-006-SCH	Revision: 2	IMR Engineering	
Date: 02/26/26	Time: 08:00	Sheet 4 of 7	Engineer: Hab S Collector	
File: SoftCore_SA_P4.SchDoc				

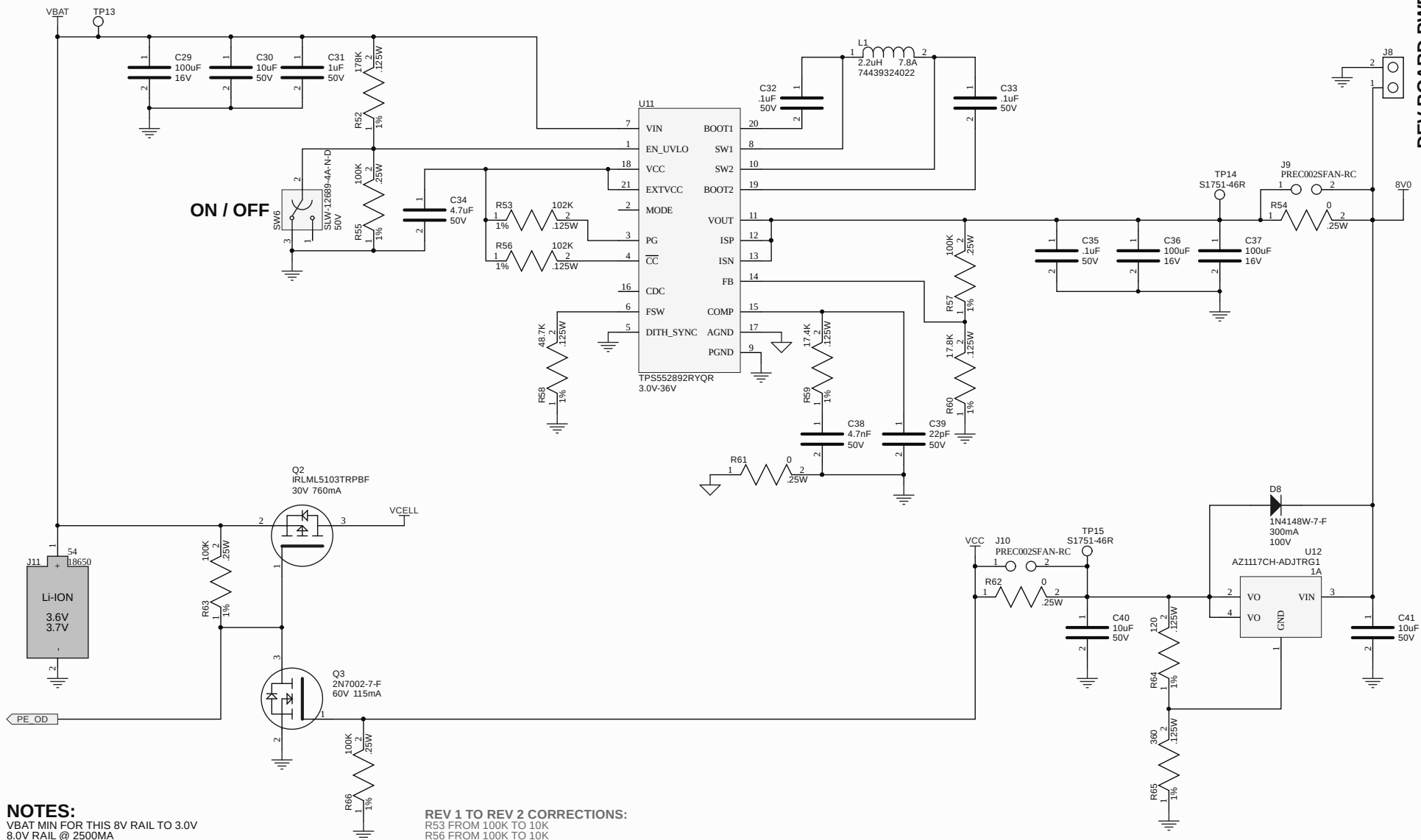


WIEN BRIDGE OSCILLATOR
 $R_w = R_f + R_{pot}$
 $C_w = C_f$
FREQUENCY @ $R_w = 11K = 4.4KHz$
FREQUENCY @ $R_w = 1K = 48.2KHz$
FREQUENCY @ $R_w = 6K = 8.0KHz$
T_SIG OUTPUT LPF -3DB @72.3KHz

NOTES:
POPULATE R JUMPER FOR AUDIO AMP PWR VCC OR 3V3 NOT BOTH

AUDIO AMP WITH VOLUME CONTROL
AUDIO SIGNAL INPUT PWM FREQUENCY = 100KHz
AUDIO INPUT FILTER SINGLE POLE @ 20.0KHz
AUDIO INPUT AC COUPLED

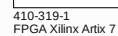
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Size: B	Number: IMR-006-SCH	Revision: 2	IMR Engineering	
Date: 02/26/26	Time: 08:00	Sheet 5 of 7	Engineer: Hab S Collector	
File: SoftCore_SA_P5.SchDoc				



Q4 FROM IRLML5103PBFC7-ND TO DMP3056-7
U13.6 FROM FLOATING TO GND CONNECT
FROM U13.13 AND U13.14 TO C43 CONNECT TO U13.13 AND U13.14 CONNECT ONLY
ADD CONNECTION U13.12 TO U13.13 AND U13.14

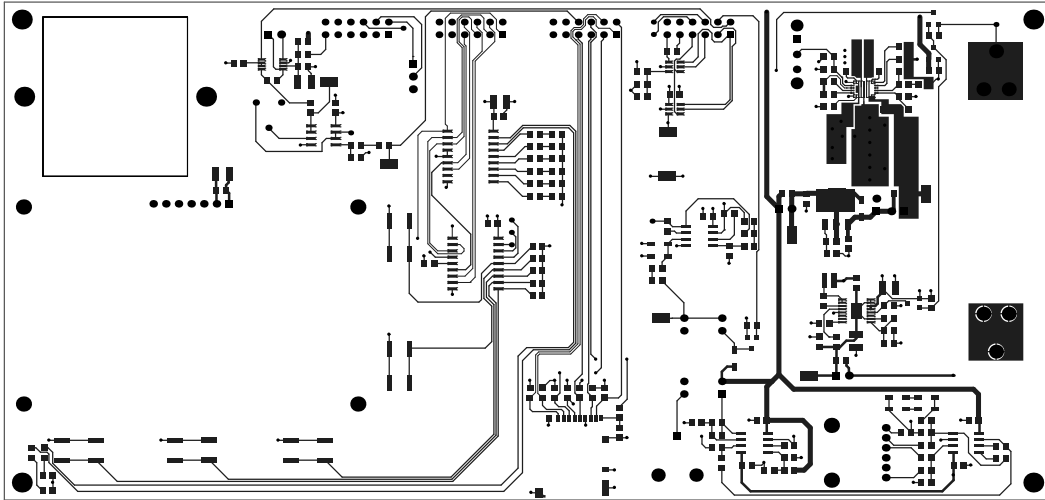


MECH9
312815-3W



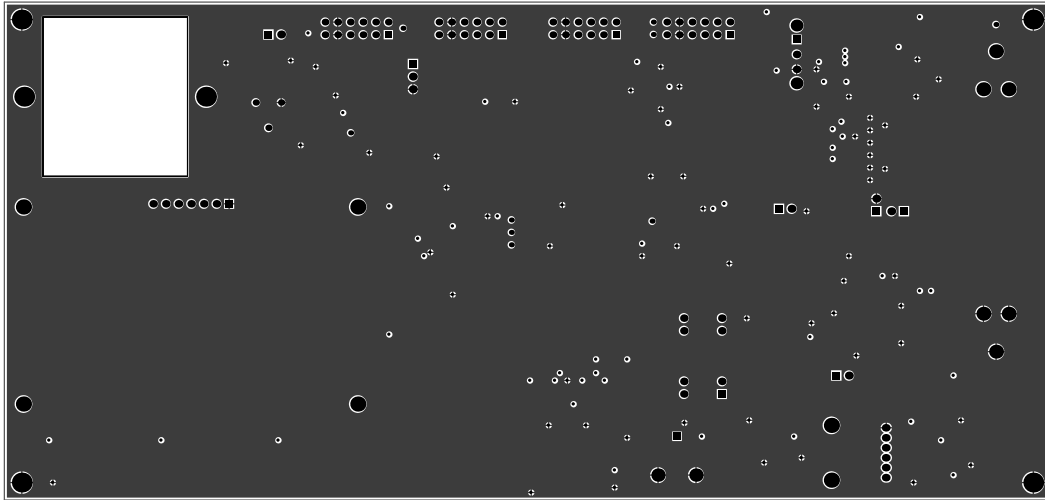
VBAT MIN FOR THIS -5V RAIL TO 3.5V
VEE RAIL -5.0V @ 200MA
VEE TURNS ON WITH VALID VCC

Title SoftCore SA Adapter			IMR Engineering 3621 Gin Way Snelville GA, 30039	IMR Engineering © 2004 IMR, Inc.
Size: B	Number: IMR-006-SCH	Revision: 2		
Date: 02/26/26	Time: 08:00	Sheet 7 of 7	Engineer: Hab S Collector	
File: SoftCore_SA_P7.SchDoc				

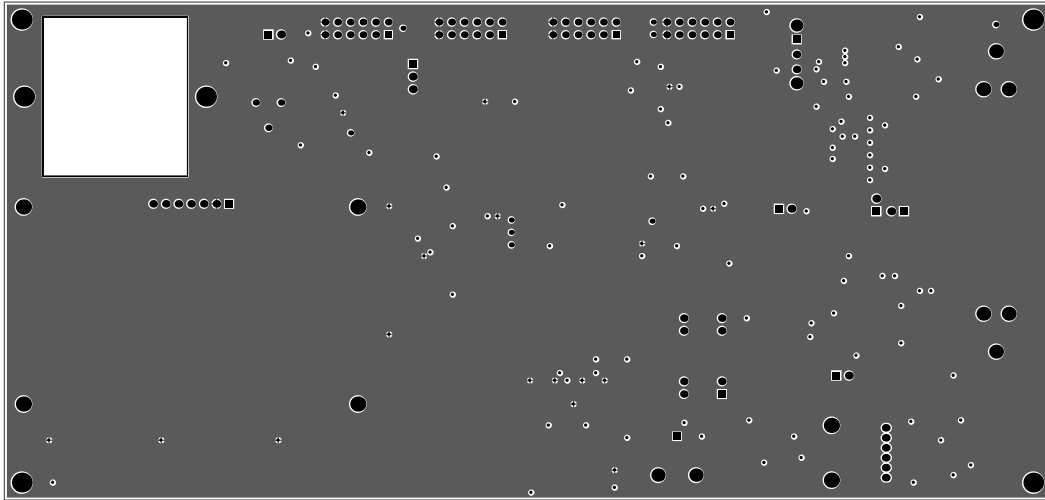


TOP

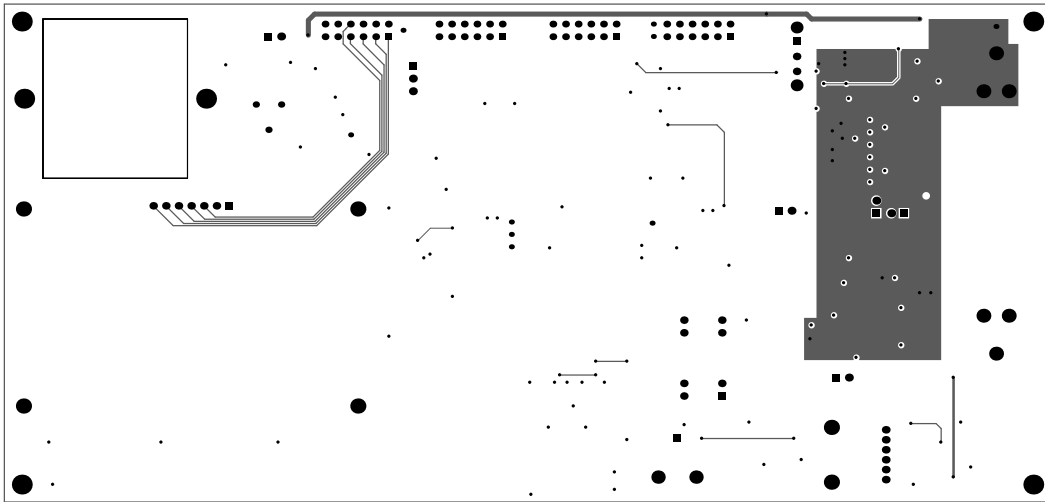
BOARD OUTLINE



SIGNAL LAYER 1
BOARD OUTLINE

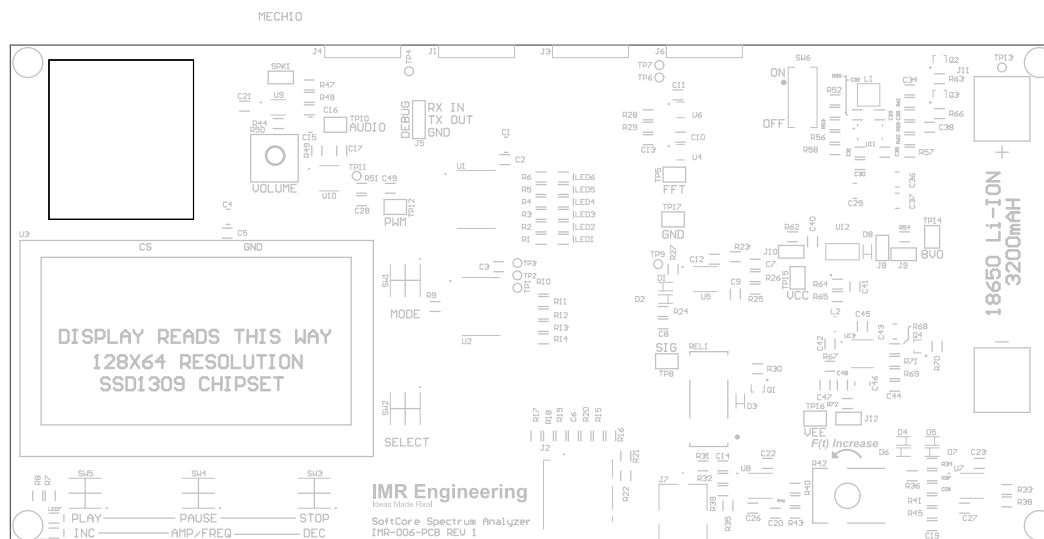


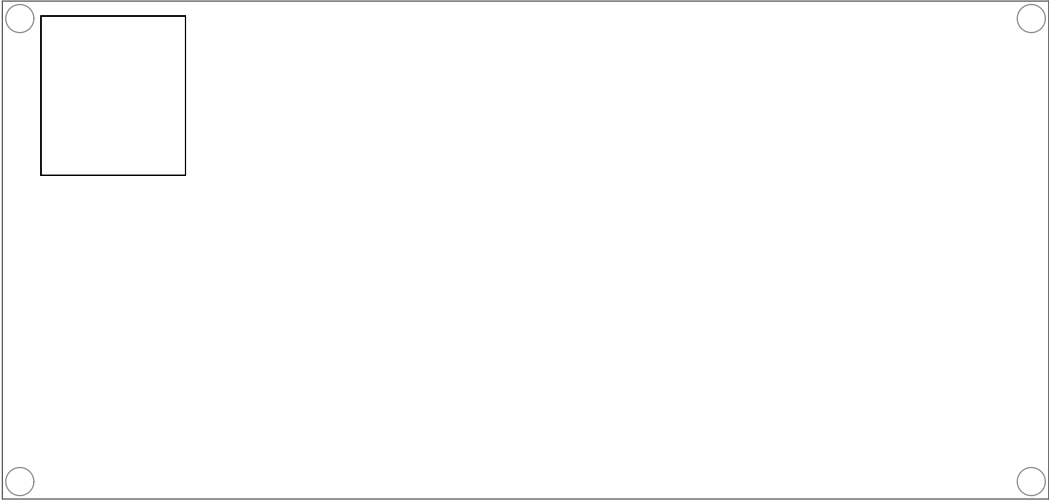
SIGNAL LAYER 2
BOARD OUTLINE



BOTTOM

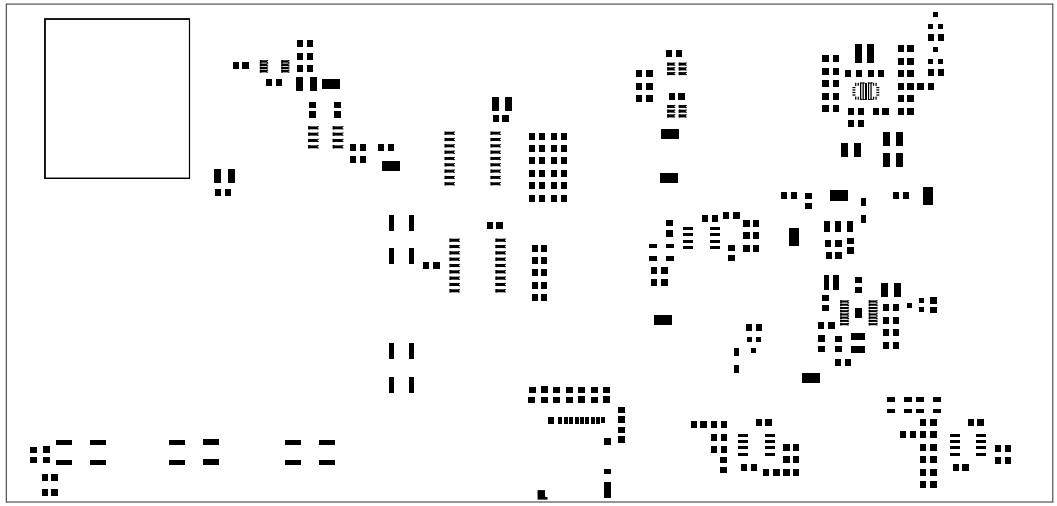
BOARD OUTLINE





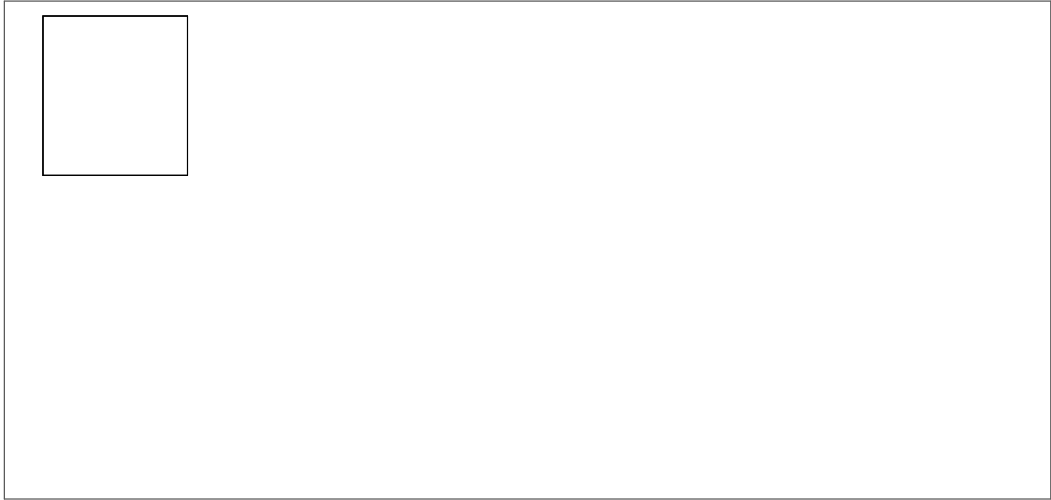
BOTTOM SILK

BOARD OUTLINE



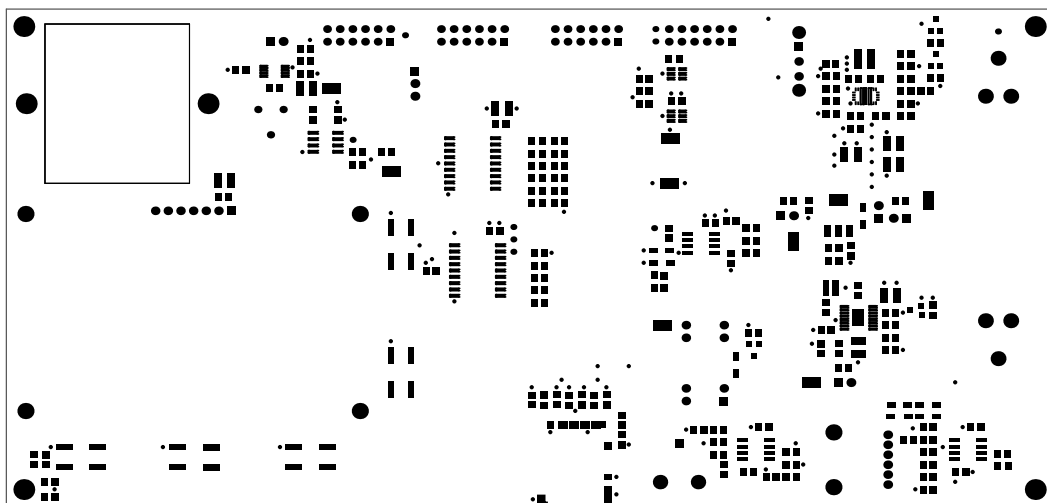
TOP PM

BOARD OUTLINE



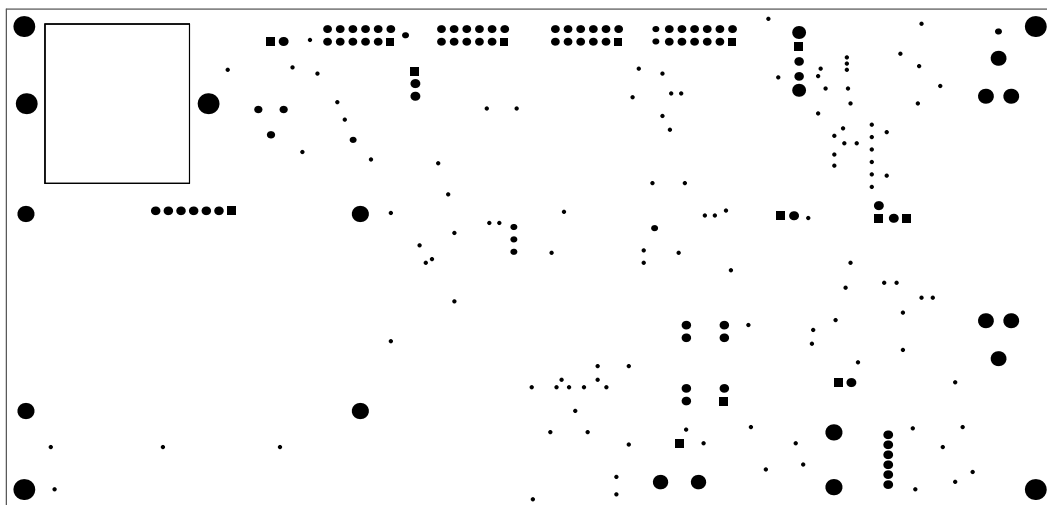
BOTTOM PM

BOARD OUTLINE



TOP SM

BOARD OUTLINE



BOTTOM SM

BOARD OUTLINE

Symbol	Hit Count	Finished Hole Size	Plated	Hole Type
o	1	0.920mm (36.22mil)	PTH	Round
o	2	1.590mm (62.60mil)	PTH	Round
*	2	1.800mm (70.87mil)	PTH	Round
x	2	2.108mm (83.00mil)	PTH	Round
x	2	3.300mm (129.92mil)	NPTH	Round
ss	3	1.010mm (39.76mil)	PTH	Round
o	4	3.048mm (120.00mil)	NPTH	Round
A	4	3.300mm (129.92mil)	PTH	Round
◇	6	1.016mm (40.00mil)	PTH	Round
▽	6	1.778mm (70.00mil)	PTH	Round
o	7	0.889mm (35.00mil)	PTH	Round
o	9	0.508mm (20.00mil)	PTH	Round
▽	11	0.762mm (30.00mil)	PTH	Round
ss	13	0.900mm (35.43mil)	PTH	Round
ss	48	1.020mm (40.16mil)	PTH	Round
□	116	0.381mm (15.00mil)	PTH	Round
236 Total				

PCB FABRICATION NOTES:

MATERIAL:

FR4 (GF per MIL-P-13949) MIN UL 94V0

Cu WEIGHT:

OUTER LAYER: 1oz

INNER LAYER: 1oz

SOLDER MASK:

TYPE: LPI (LIQUID PHOTO-IMAGE)

COVER: SMOBC (SOLDER MASK OVER BARE Cu)

COLOR: RED (PROTOTYPE) OR GREEN (PRODUCTION) GLOSS

OVERALL PCB THICKNESS:

PCB: 1.6mm (63MIL)

TOLERANCE: 0.178mm (7MIL)

PCB ELECTRICAL TEST

TESTED TO GERBER DATA

PURCHASE ORDER TO OVERRIDE

HOLE DIAMETER TOLERANCE:

PLATED HOLE TOLERANCE: 0.076mm (3MIL)

NON PLATED HOLE TOLERANCE: 0.076mm (3MIL)

SILK SCREEN

SIDES: TOP

COLOR: WHITE

TYPE: LPI (PREFERRED)

SURFACE FINISH:

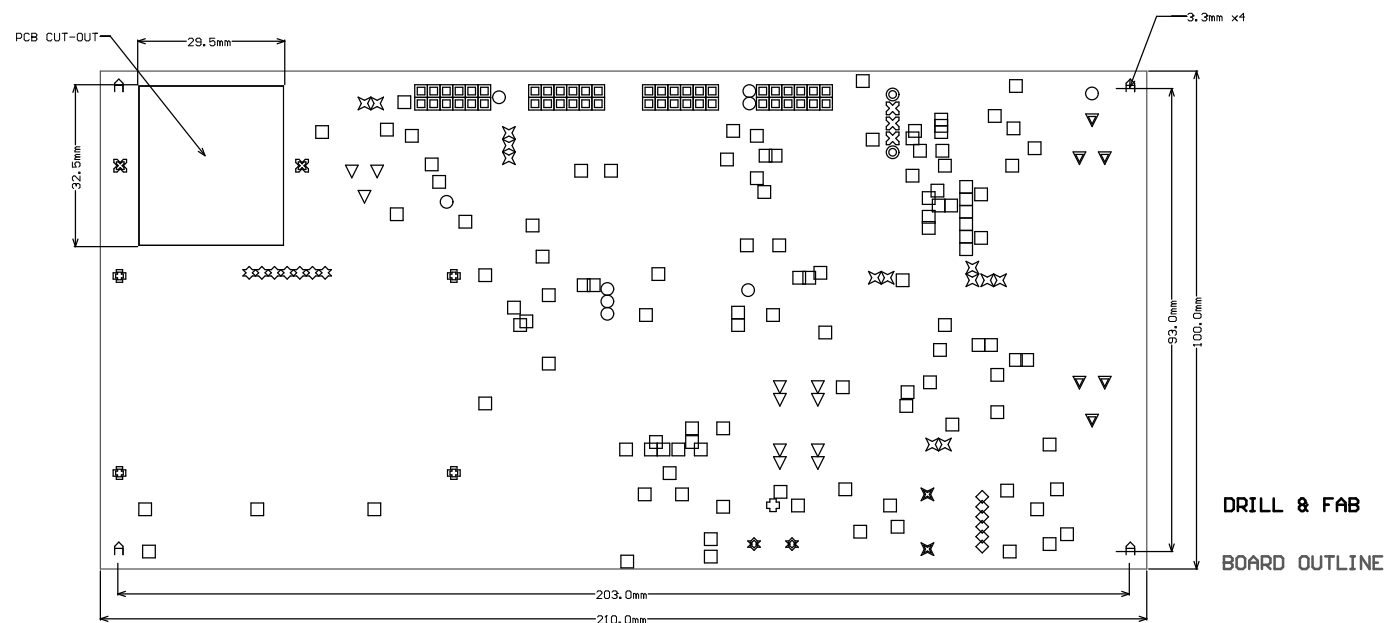
PROTOTYPE: HASL OR ENIG

PRODUCTION: ENIG (ONLY IF SPEC ON PURCHASE ORDER)

ACCEPTABILITY:

STANDARD: IPC-A-600 (LATEST REV)

MFG TO ADD: DATE CODE, UL FLAME CODE



IMR Engineering
SoftCore SA

REV 1
02/26/2026

ENGG: IMR-006-PCB
Hab Collector

